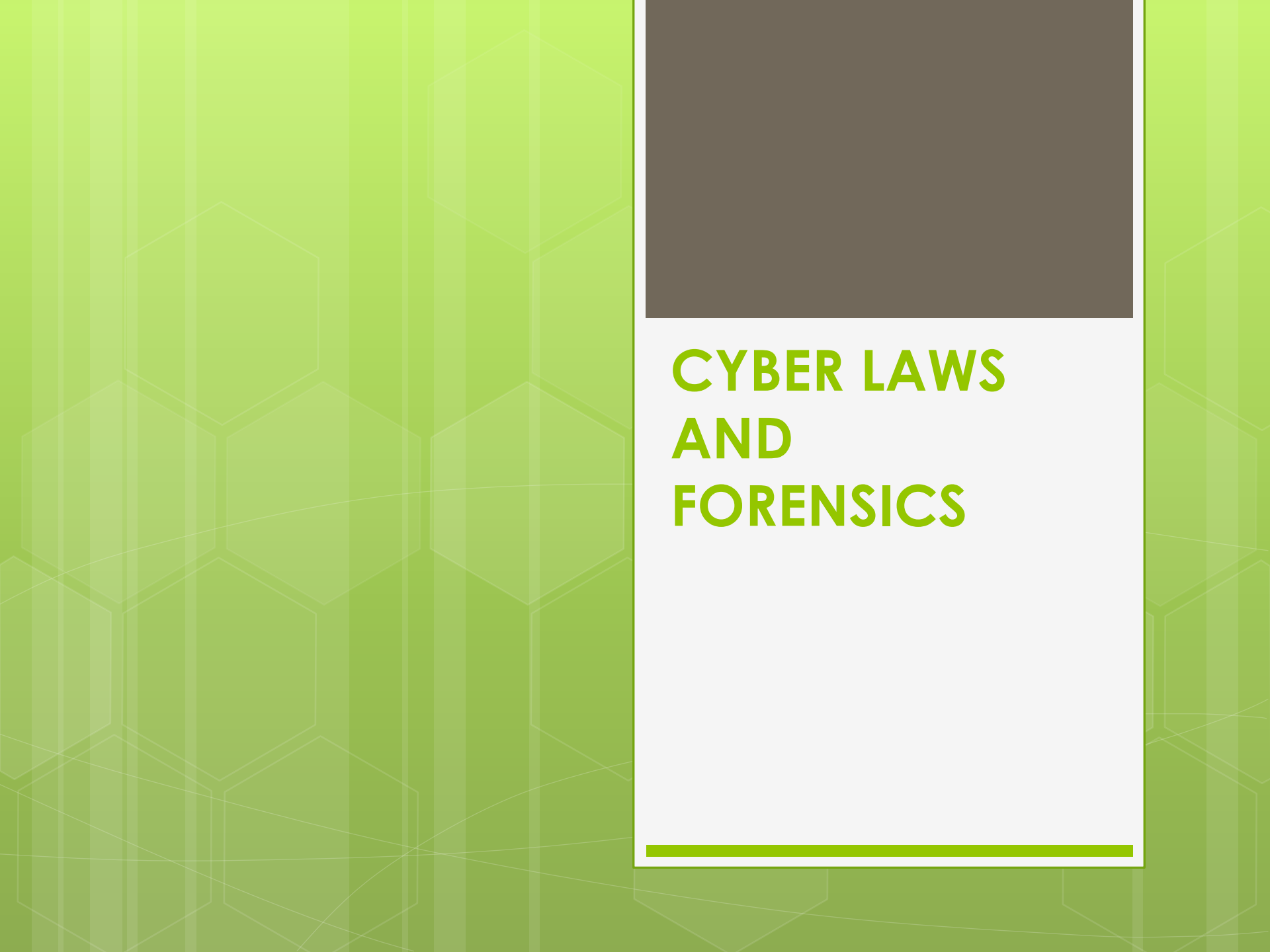




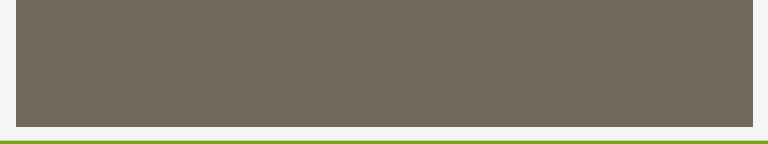
CYBER LAWS AND FORENSICS

- CO1 To understand the information technology act.
- CO2 To understand various penalties related to cybercrimes.
- CO3 Explain a computer crime and the concept of rules or policy violations.
- CO4 Gather evidences and preserve the collected evidence with the required knowledge on various storage format choices.
- CO5 Prepare forensics reports both using tools and manually and explain ethics and code for expert witness.

Introduction to Cyber Law

Cyber law, also known as internet law or IT law, refers to the legal framework that governs activities conducted on the internet and digital platforms. It is essential to regulate cyberspace and protect individuals, businesses, and governments from cyber threats, fraud, and other digital crimes.

Cyber laws are crucial in the digital age to protect individuals and organizations from various online threats, ensure the integrity of online transactions, and safeguard intellectual property. They also provide a legal framework for e-commerce and data protection, fostering trust and encouraging innovation in the digital realm.



With the rapid growth of digital technology, online transactions, and social media, cyber law has become crucial to ensure security, privacy, and ethical use of the internet.

Needs or Importance of Cyber Laws

1. Protection Against Cyber Crimes

- The internet has enabled various forms of cyber crimes, such as:
- **Hacking** – Unauthorized access to computer systems and networks.
- **Identity Theft** – Stealing personal information for fraudulent activities.

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- **Phishing** – Deceptive attempts to obtain sensitive information.
 - **Cyber Stalking & Harassment** – Online threats, bullying, and defamation.

Cyber laws provide legal remedies and penalties to prevent and punish cybercriminals.

2. Securing Digital Transactions & E-Commerce

With the rise of online banking, e-commerce, and digital payments, cyber laws ensure:

- Secure transactions and protection against fraud.
- Regulations for online business contracts and digital signatures.
- Consumer rights protection in online purchases.

For example, the **Information Technology (IT) Act, 2000 (India)** and **General Data Protection Regulation (GDPR, EU)** help protect users from financial fraud.

3. Data Protection and Privacy

- Personal and sensitive data is constantly shared online through social media, emails, and cloud storage.
- Companies and organizations collect user data, making it essential to regulate data handling and prevent misuse.
- Laws like **GDPR (Europe)** and **Personal Data Protection Bill (India)** ensure individuals have control over their personal information.

4. Intellectual Property Rights (IPR) Protection

Cyber laws help protect digital content such as:

- **Copyrighted material** – Prevents unauthorized use of music, videos, articles, and software.
- **Patents & Trademarks** – Protects innovations, domain names, and branding.

For example, **Digital Millennium Copyright Act (DMCA, USA)** helps combat online piracy and copyright infringement.

5. Preventing Cyber Terrorism and National Security Threats

- Terrorists and criminals use the internet for illegal activities, including spreading propaganda, recruiting members, and cyber-attacks.
- Cyber laws enable governments to monitor, investigate, and take action against cyber terrorism.

Acts like **The IT Act (India)**, **Cybersecurity Information Sharing Act (USA)** allow authorities to combat cyber threats effectively.

6. Regulating Social Media and Fake News

- Cyber laws address issues related to fake news, hate speech, and misinformation spread on social media platforms.
- Social media regulations prevent abuse, protect individuals, and hold platforms accountable for content moderation.
- Laws like **The Information Technology (Intermediary Guidelines) Rules, 2021 (India)** mandate social media platforms to take responsibility for content moderation.

Cyber law-historical perspectives

Cyber law, encompassing regulations related to computer networks and cyberspace, has a complex history marked by evolution and adaptation. Early developments focused on data protection and computer fraud, while the internet boom and the rise of the digital age spurred a broader need for cyber law.

1 Early Developments (1970s-1980s):

As computer networks began to interconnect, initial considerations for legal protections and responsibilities started to surface.

The Computer Fraud and Abuse Act of 1986 in the United States marked one of the first legislative attempts to address cyber-related crimes.

2. The Internet Boom (1990s):

- The rapid growth of the internet in the 1990s raised new legal challenges, particularly regarding online communication, e-commerce, and digital signatures.

3. Evolving Concerns and Adaptations:

- As technology advanced, so did cybercrime, leading to amendments to the Information Technology Act to address new forms of cyber offenses and strengthen data protection.
- The rise of data privacy concerns spurred the development of regulations like the General Data Protection Regulation (GDPR) in the EU.

4. Global Perspective:

- Cyber laws are not limited to individual nations and are increasingly addressing the global nature of cyberspace.
- International conventions and agreements are being developed to promote cybersecurity and cooperation among countries in addressing cybercrime.

5. Current Trends and Challenges:

- The focus of cyber law is shifting towards emerging technologies and addressing the legal challenges they present.
- Challenges remain in adapting cyber laws to a rapidly evolving digital landscape while ensuring the protection of individual rights and fostering innovation.

Cyber Space

Cyberspace is the virtual space created by interconnected computer networks and the internet. It's where digital information, communication, and interactions take place.

Cyberspace includes the Internet, telecommunications networks, computer systems, and other networked devices.

It enables communication, information access, online shopping, social media engagement, and a vast array of other digital activities.

Cyberspace is a critical component of modern life, supporting global communication, economic transactions, critical infrastructure, and technological innovation. Cyberspace also presents security challenges, such as cyberattacks, data breaches, and digital espionage.

Deception by squatting in cyberspace

Cybersquatting is also known as domain squatting, which involves registering domain names that are identical or confusingly similar to existing trademarks, company names, or personal names.

Cyber squatters often register domain names with the intention of deceiving internet users into thinking they are accessing the legitimate website, leading to potential fraud, data theft, or other harmful consequences.

Types of Cybersquatting:

Typosquatting: Registering domain names with slight misspellings of well-known brand names (e.g., "Goggle.com" instead of "Google.com").



Name Jacking: Registering domain names that are personal names of famous people or executives.

Brand Jacking: Using variations of well-known brand names to phish or impersonate.

Combo Squatting: Adding words to well-known brands (e.g., "nike-outlet-sale.com").

Examples of Deception through Cybersquatting:

Creating Fake Websites:

Cybersquatters may create clone websites of legitimate businesses, deceiving users into trusting these fake sites and potentially leading to phishing attempts or data theft.

Damaging Brand Reputation:

Cybersquatters may use the domain name to negatively impact the reputation of the genuine brand, for example, by hosting negative content or misinformation.

protection of copyright on cyberspace

Protection of copyright in cyberspace is a critical aspect of digital rights management and intellectual property law, especially given the ease with which content can be copied, shared, or modified online.

how copyright is protected in cyberspace

1. Legal Frameworks

Legal frameworks in cyberspace refer to the body of laws, regulations, and standards that govern online activities and protect users from cybercrimes and threats.

Data Protection and Privacy

Cybercrime Prevention

E-commerce Regulation

Intellectual Property Protection

Major Legal Frameworks:

Information Technology Act, 2000 (India):

This act is a foundational legal framework in India, addressing cybercrimes, e-commerce, and digital signatures.

General Data Protection Regulation (GDPR):

The GDPR is a key regulation for data protection in the EU, setting standards for how personal data is handled.

2. Challenges in Cyberspace

Ease of Copying and Sharing Digital content (music, videos, software, books) can be copied with no degradation in quality.
Anonymity and Jurisdiction Infringers can hide behind anonymity or operate from countries with weak IP enforcement.

3. Methods of Protection

Digital Rights Management (DRM) Technological tools that restrict copying or sharing of digital content (e.g., encryption, access control). **Watermarking** Digital watermarks can embed ownership information into the content, which can be traced if redistributed. **Content ID Systems** Platforms like YouTube use automated systems to detect and manage copyrighted material.

4. Enforcement and Regulation

Monitoring and Enforcement Tools Use of bots, crawlers, and AI to monitor infringement. **Court Orders and ISP Blocking** Courts can order ISPs to block access to infringing websites. **Civil and Criminal Penalties** Penalties for copyright infringement can include fines and imprisonment.

5. Ethical and Educational Measures

Awareness Campaigns: Educate users about respecting intellectual property. Institutional Policies: Universities and corporations may adopt policies promoting legal use of digital content.

Infringement of copyright on cyberspace

Copyright infringement in cyberspace, or online, involves the unauthorized use of copyrighted material on the internet. This can take various forms, including illegally reproducing, transmitting, linking to, framing, or using copyrighted works online. Examples include downloading music or movies illegally, uploading copyrighted content without permission, or using unauthorized copies of software.

Specific Examples of Copyright Infringement in Cyberspace:

1. Illegal Downloading:

Downloading copyrighted music, movies, video games, software, or e-books without permission is a common form of infringement.

2. Unauthorized Uploading:

Uploading copyrighted material to the internet without permission, such as sharing music or videos on social media or online platforms, is also infringement.

3. Linking and Framing:

Linking to or framing copyrighted content without permission can also constitute infringement, especially if the linked content is being reproduced.

3. Software Piracy:

Downloading or distributing unauthorized copies of licensed software, or using cracked versions, violates copyright laws.

4. Plagiarism:

Copying and pasting text, images, or other content from websites without giving proper credit is a form of infringement.

Caching and Archiving:

Unauthorized caching or archiving of copyrighted materials without permission can also be infringement.

Linking, hyperlinking and framing

Linking and framing are essential web technologies, and hyperlinking is a type of linking. Linking connects web pages or websites, while framing displays content from one website within another. Hyperlinking, often in the form of a link or button, directs users to another page, while framing divides a web browser into multiple areas for content from different sources.

Linking and Hyperlinking:

Linking: is the general term for connecting web pages or websites, enabling users to navigate between different online resources.

Hyperlinking: (also called a hyperlink) is a specific type of link that often appears as a clickable text or image on a webpage.



Surface linking: Links to a website's homepage.

Deep linking: Links directly to a specific page within a website.

Framing:

Framing: is the process where a website displays content from another website within its own page, but within a separate frame.

This allows users to view content from different sources without leaving the original page.

Frames can be used to create different sections or areas on a webpage, displaying content from various websites.

The original source is shown, but it appears as part of your webpage, often misleading users into thinking the content is yours.

ISP in cyberspace

An ISP, or Internet Service Provider, is a company that provides internet access to individuals and businesses. They connect users to the internet, enabling them to browse, email, and use other online services.

Besides internet access, ISPs may offer other services like email, web hosting, and domain registration.

Users pay ISPs for the internet access and other services they provide.

ISPs provide the infrastructure (wires, cables, satellite connections, etc.) and the means to connect to the internet.

ISPs invest in and maintain the network infrastructure that allows users to access the internet.



Examples of ISPs:

Airtel, ACT Fibernet, TechTarget says

Comcast, Cox Communications, and CTS Telecom are also listed as examples

Other local and regional ISPs.

cyberspace and protection of patents in India

In India, patent protection in cyberspace, especially for cybersecurity innovations, is a complex issue. The Indian Patents Act, 1970, allows for patenting innovations that involve software with a "technical effect," meaning they have a demonstrable impact beyond simply being software code. This can include cybersecurity inventions that enhance security or efficiency in unique ways.

Software and Cyberspace:

While software itself is generally not patentable, inventions involving software that provide a tangible "technical effect" can be patented.

Cybersecurity Innovations:

Many cybersecurity inventions, including those related to online security and data protection, can qualify for patent protection under this "technical effect" criterion.

Challenges:

The nature of cyberspace and the ease of digital copying present challenges in protecting intellectual property, including patents.

IT Act Support:

The Information Technology Act, 2000, complements patent and other IP laws by addressing issues like data theft and hacking, which are relevant in the context of cyberspace.

Blockchain and IoT:

Blockchain technology and the Internet of Things (IoT) are areas where further legal frameworks may be needed to address intellectual property protection in the evolving digital landscape.

Cyber security involves a range of tools and strategies to defend digital spaces from cyber attacks. Patent law grants inventors exclusive rights over their unique cyber security solutions, which encourages:



Research and Development: Patent protection motivates individuals and companies to invest in new cyber security technologies.

Innovation: With patent protection, inventors can safely develop advanced solutions, knowing their efforts are protected.

Commercialisation: Patents enable inventors to monetise their innovations, helping spread cyber security technologies across industries.

Key Sections in Patent Law for Cyber Security

Section 3(k): This section excludes “computer programs per se” from patentability. However, if a cyber security solution has a technical effect or is linked to hardware, it may be patentable.

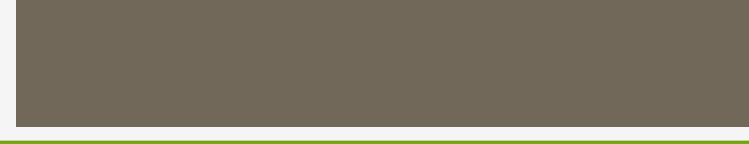


Section 39: Innovations with defense or security applications require approval before any international patent filings, safeguarding sensitive technologies.

Section 10: Patent applications must provide enough technical detail for replication, striking a balance in cyber security between protection and confidentiality.

Steps to Secure a Patent in Cyber Security

1. **Conduct a Comprehensive Patent Search:** Ensure your invention is unique by conducting a prior art search.
2. **Develop a Detailed Patent Application:** Cyber security inventions require detailed applications, especially when they involve software. Clearly explain how your innovation achieves a technical effect.
3. **Submit the Application:** File with the Indian Patent Office or, if seeking international protection, through the Patent Cooperation Treaty (PCT).



4. Undergo Examination: The patent office will examine the application, during which inventors must respond to any queries to satisfy all criteria.

5. Receive the Patent Grant: Once approved, the inventor receives a patent, usually valid for 20 years, securing exclusive rights in India.